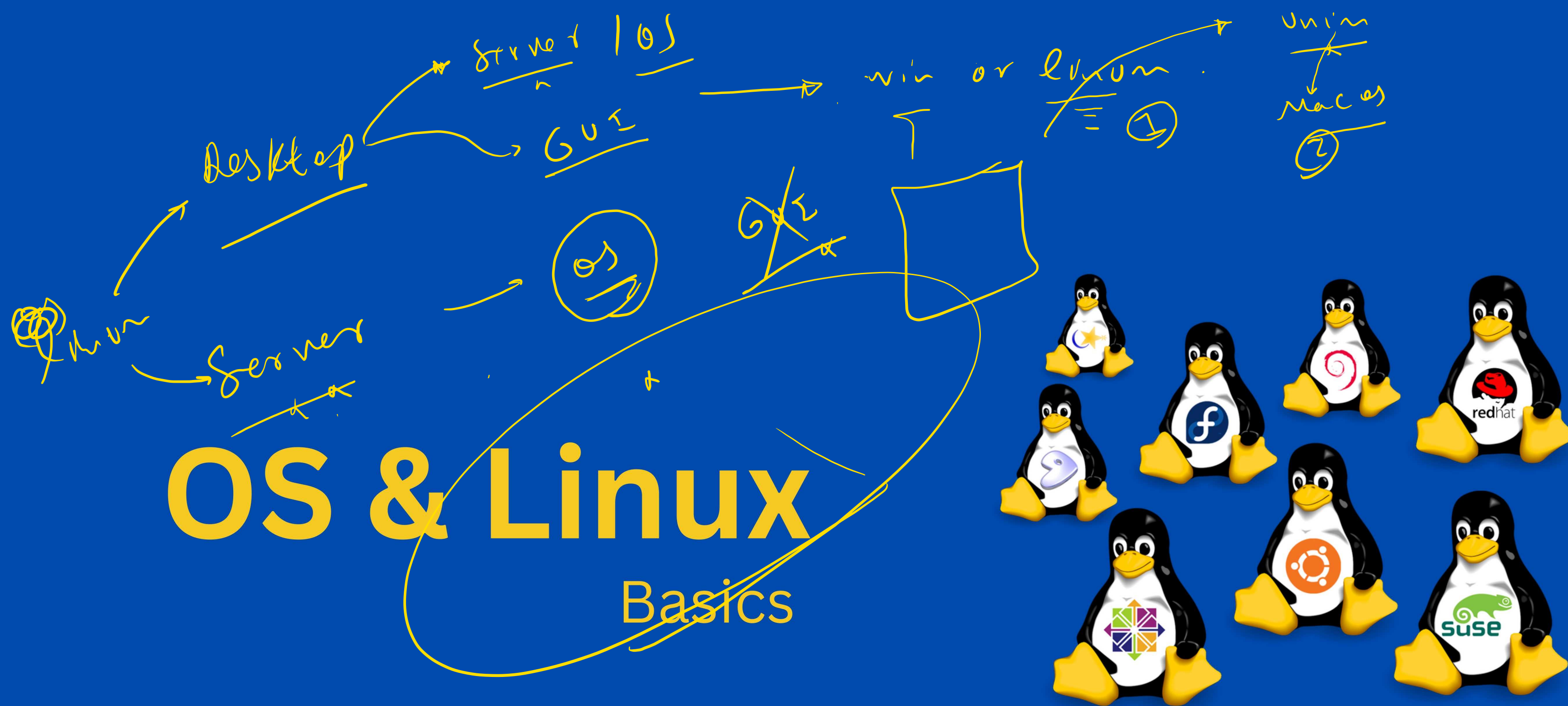




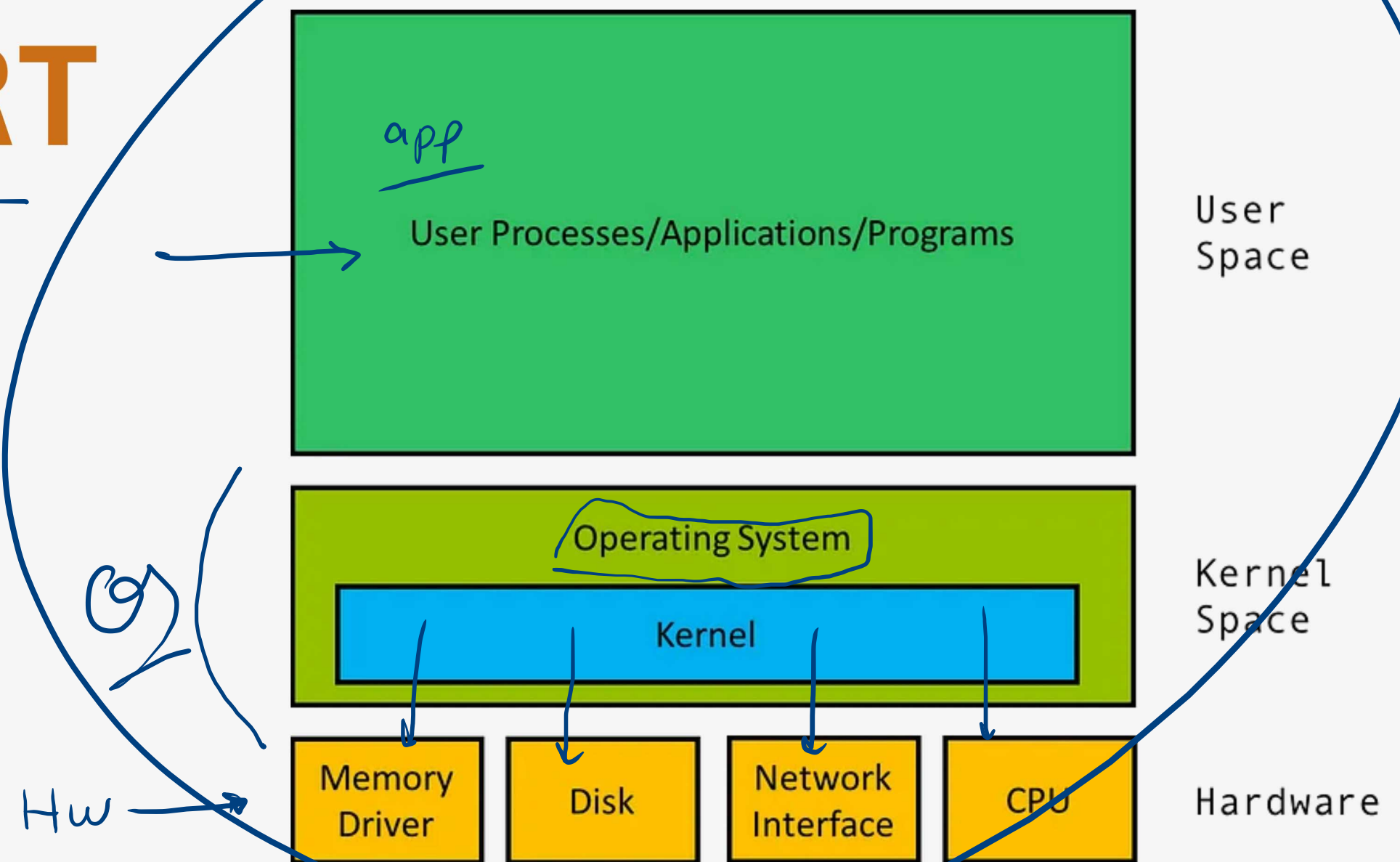
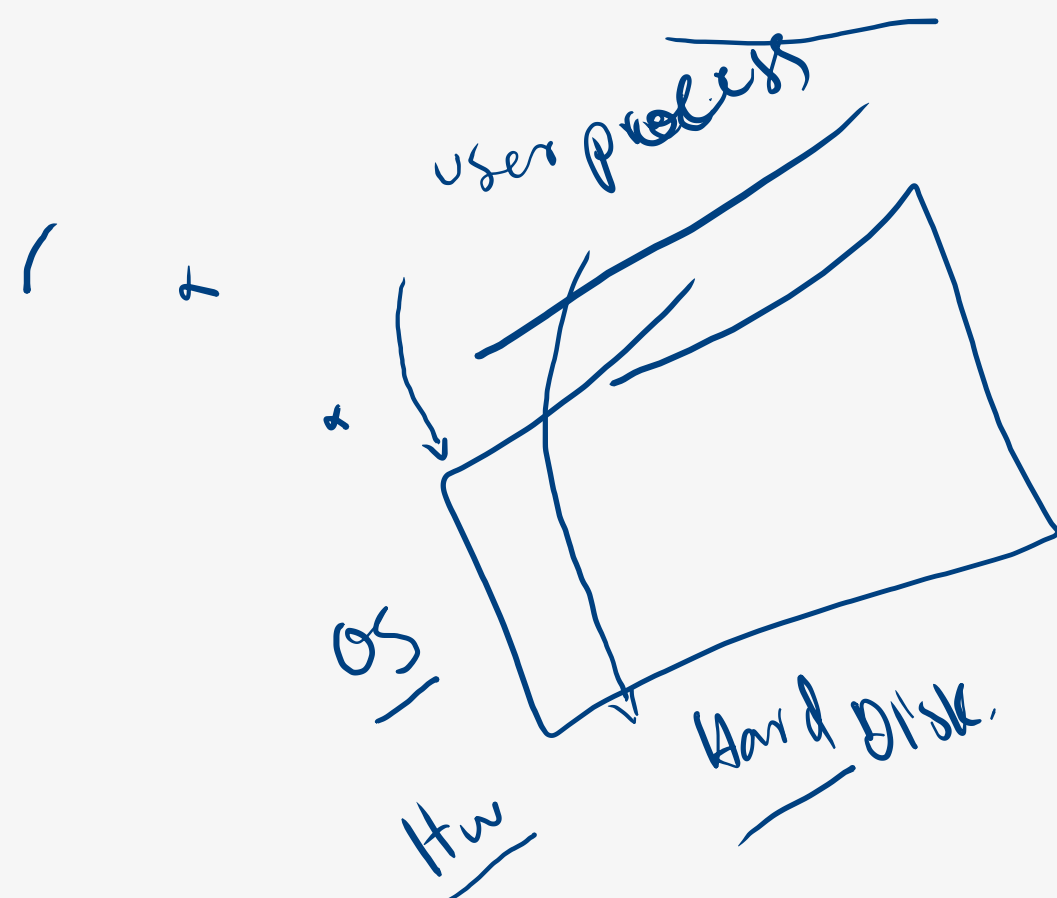
What is Linux?

Linux is **not** an Operating
System (OS), It's a **Kernel**.

What is a Kernel?

- It is the main layer between the OS and computer hardware.
- It helps with tasks such as process, memory management, and file systems.

- It's the **HEART** of the OS



OS $\longrightarrow$ vountu $\longrightarrow$ 5.96B

kernel $\longrightarrow$ linux $\longrightarrow$ 150 MB



Linux kernel

OS

C++

سخت افزار

OS
Raspbian

ARM

Raspberry

embeded linux

ARM
300MHz

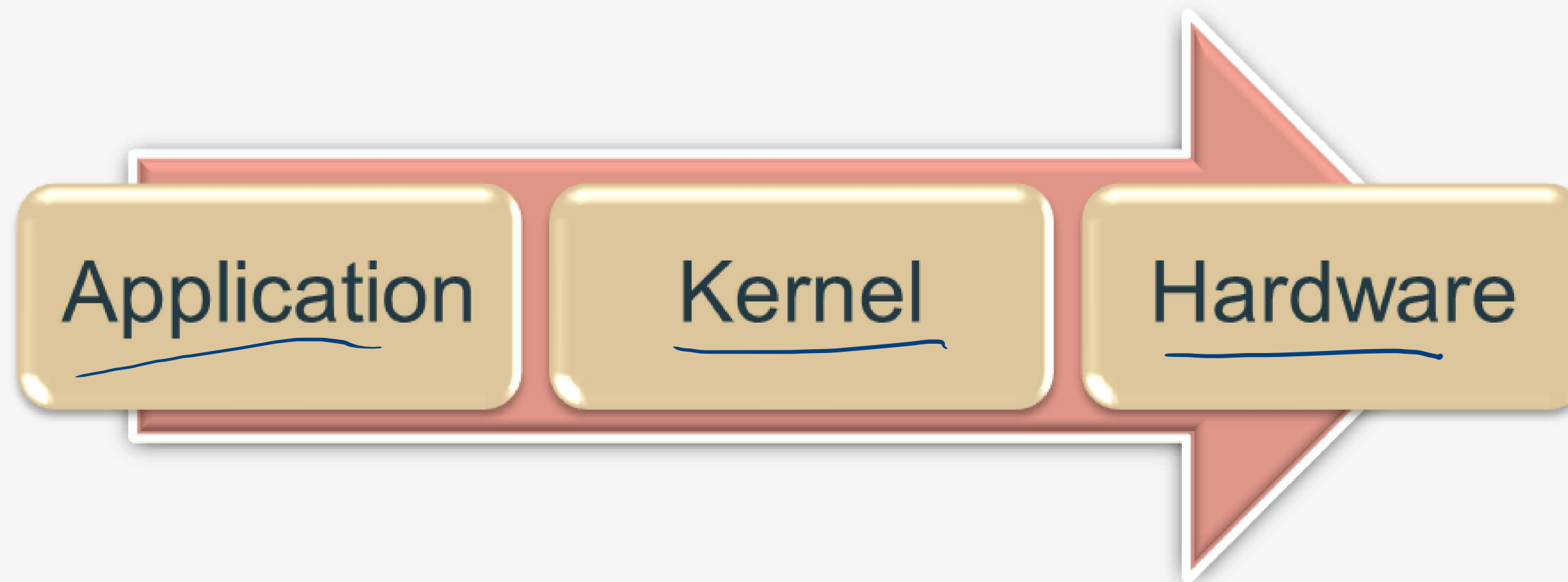
Board 1G

DRAM → 1GB
→ USB

OS - Arria OS
Linux + OS

Linux kernel → 150 MHz

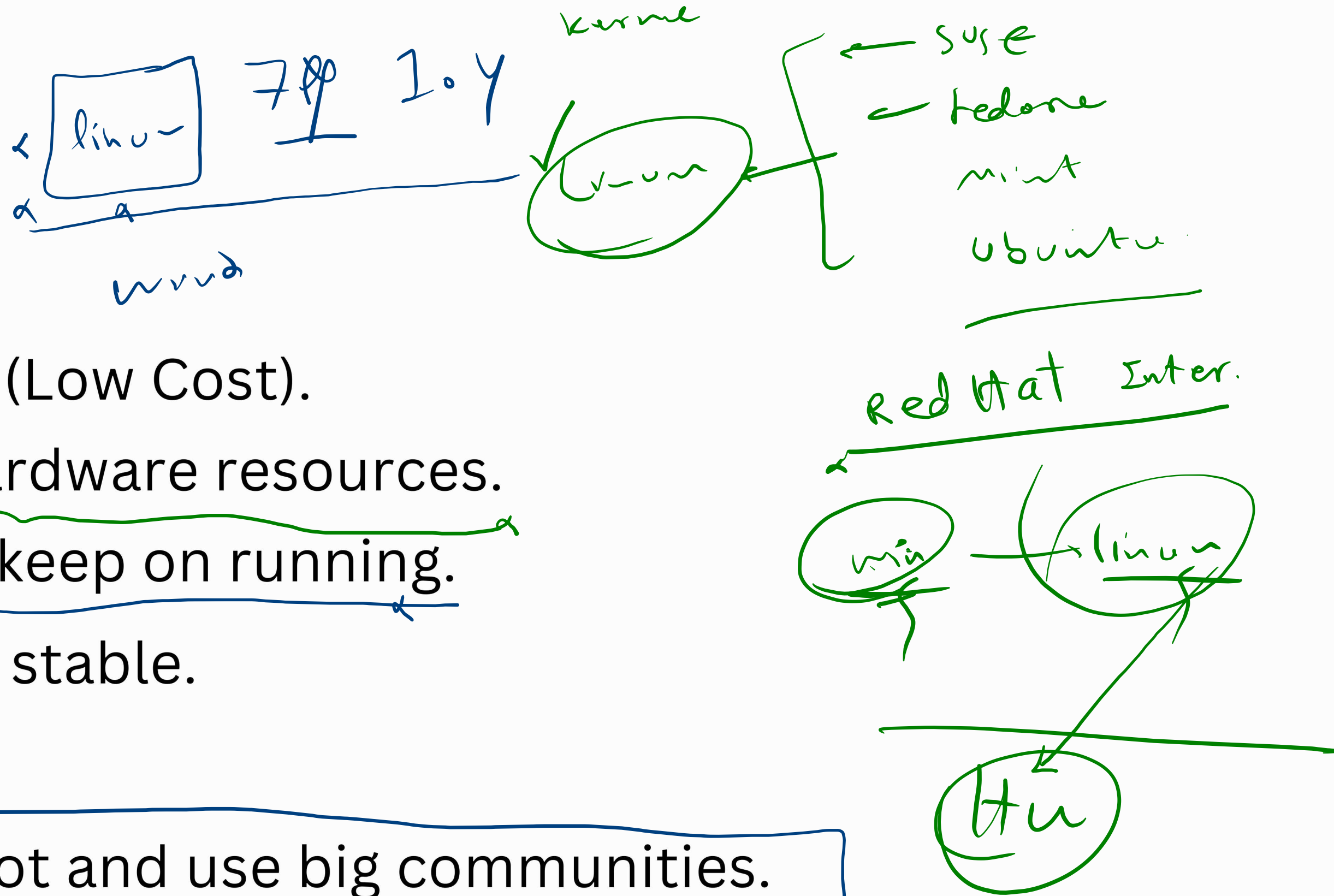
C++ / C



What is Linux Kernel?

- Kernel = The core of all Linux operating systems.
- The kernel is essentially the communication layer between the operating system and the hardware.
- The Linux kernel is built exactly on the code written by Linus Torvalds.

Why Linux



- Linux is free to use (Low Cost).
- Require minimal hardware resources.
- Linux was made to keep on running.
- Linux is secure and stable.
- Linux is scalable.
- Easy to troubleshoot and use big communities.

- Over +1000 Distributions.



multimedia

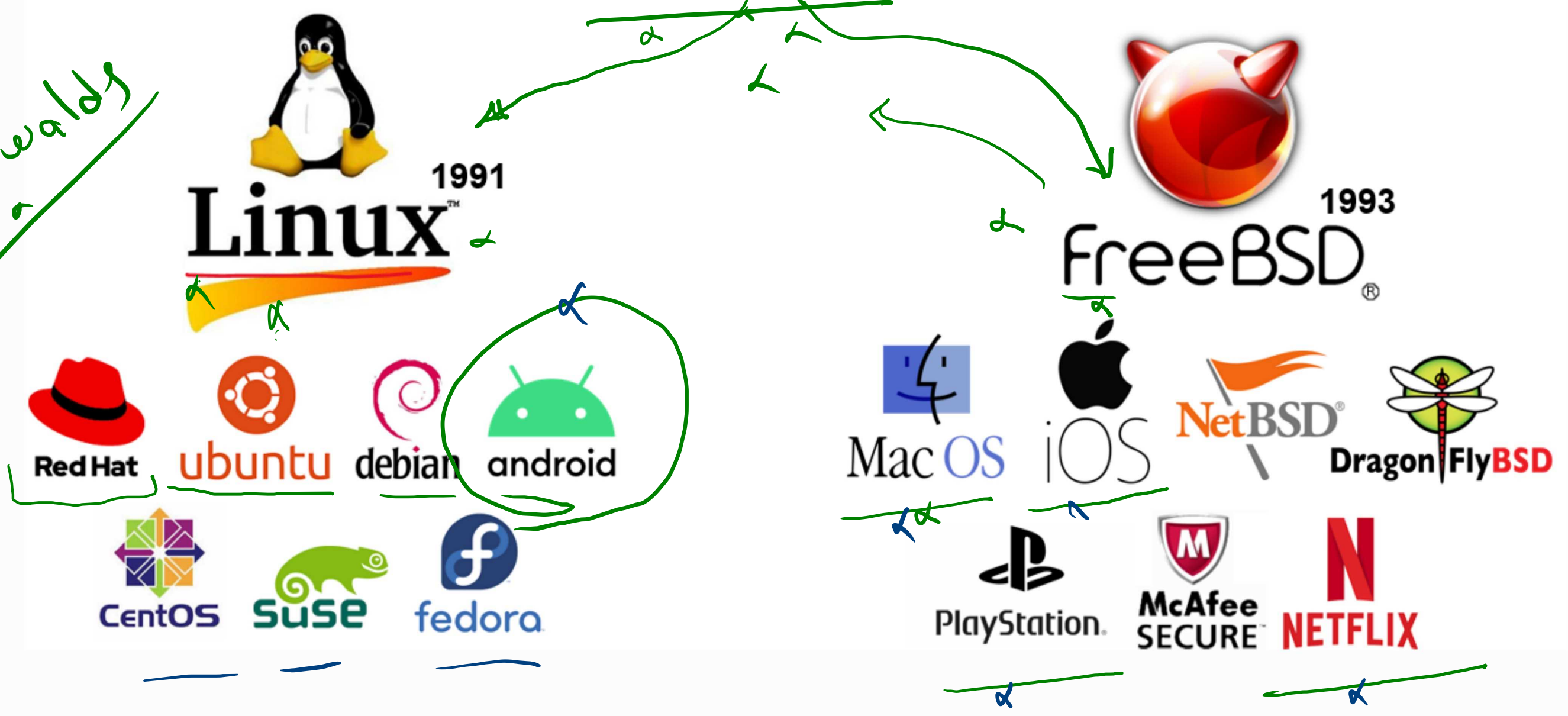
Android

linux → OS : Android

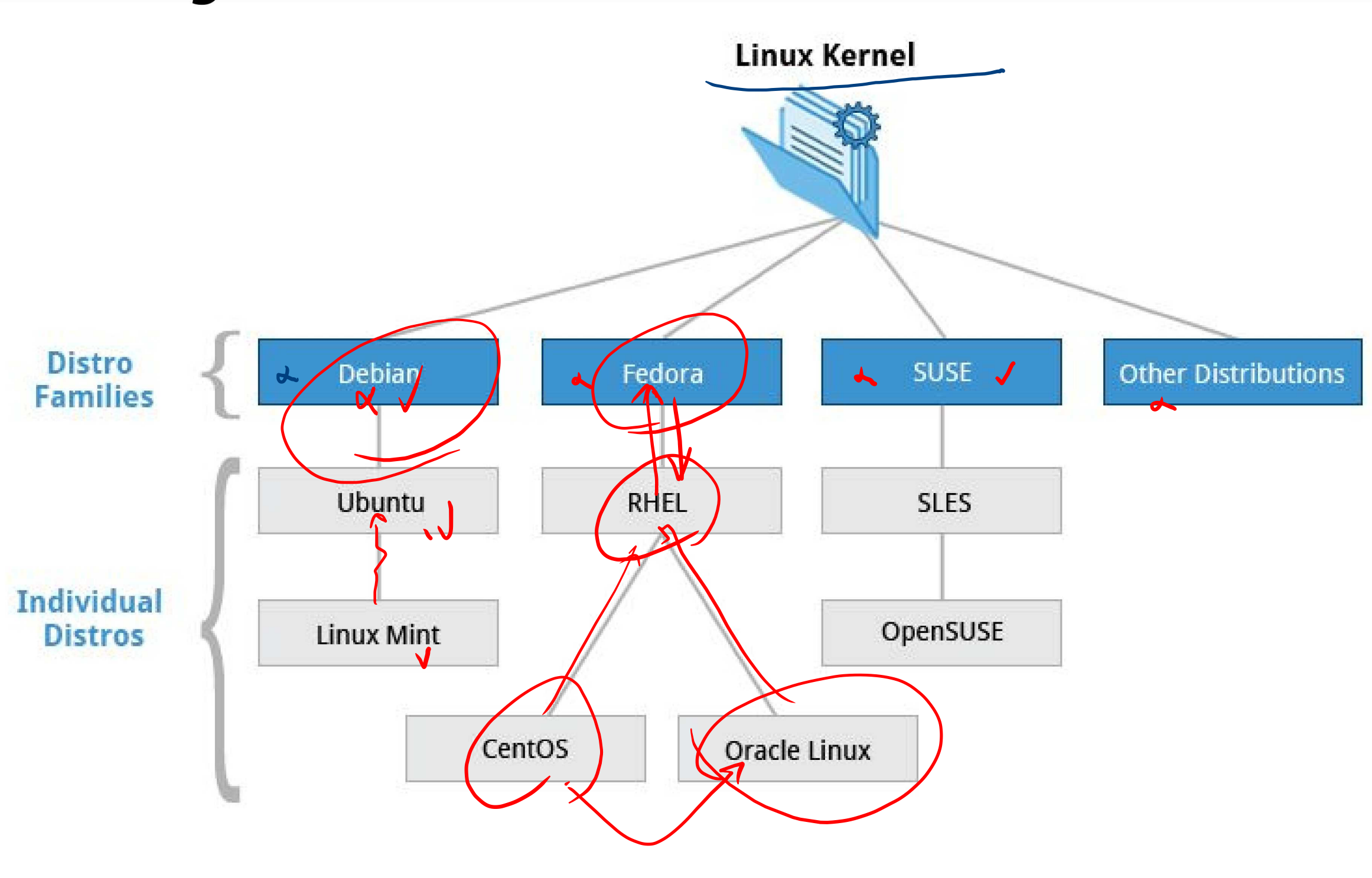
Bmw

UNIX

Derivates



Linux Family



Linux User Interfaces (UIs)



There are generally two user interfaces for Linux management:

GUI

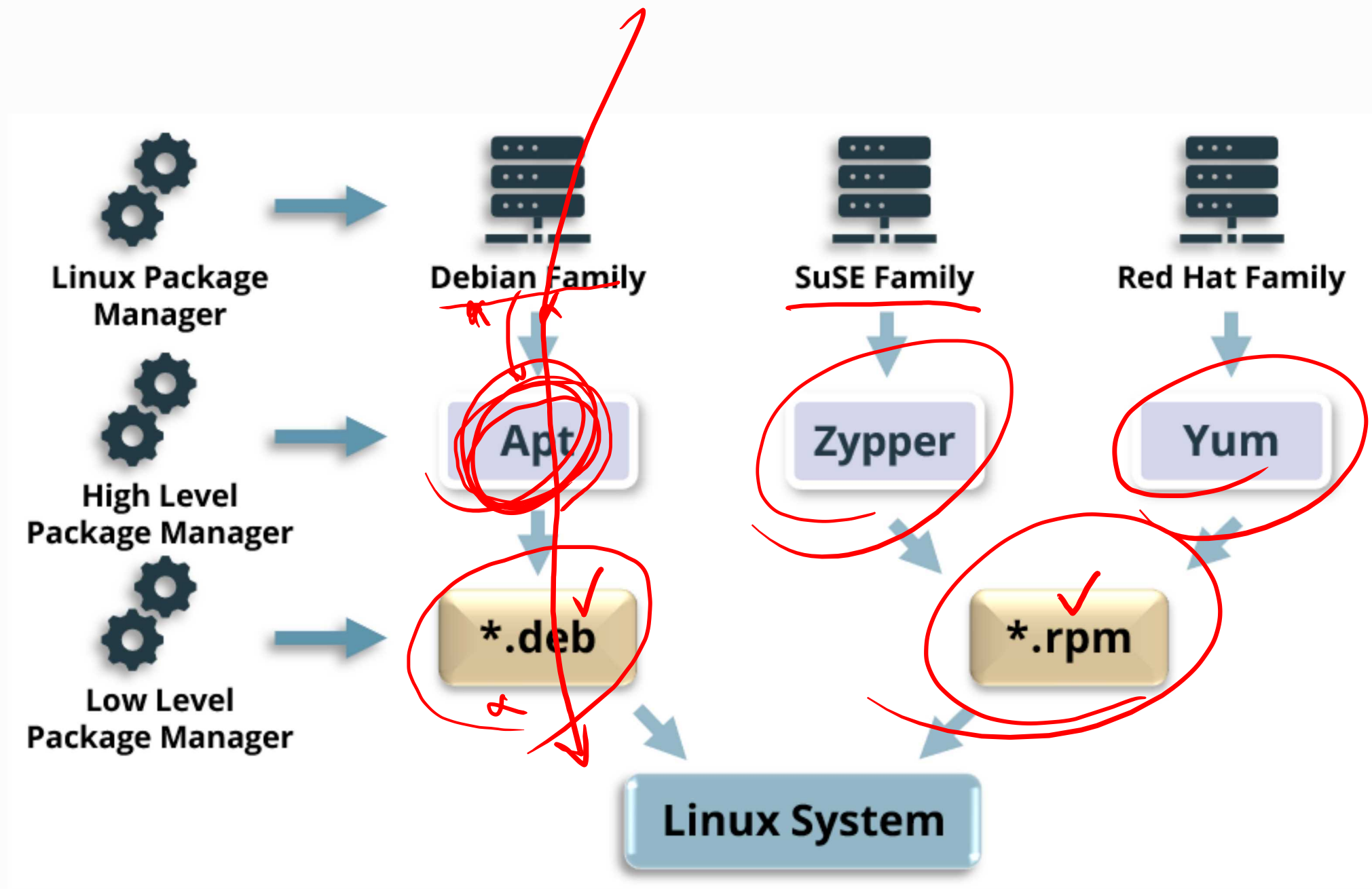
Graphical User Interface

CLI

Command Line Interface

Linux Software Packages

Repository is a collection of software and can be used by various Package Managers.



Installing Software from Repository

- Software installation on Red Hat Family is done using YUM (Yellowdog Updater Modified).
- Software installation on Debian Family is done using APT (Advanced Packaging Tool).
- How to install and uninstall software on Linux:

- **In Red Hat Family:**

[user@host ~]\$ `yum install/remove cowsay`

- **In Debian Family:**

user@host:~\$ `apt-get install/remove figlet`

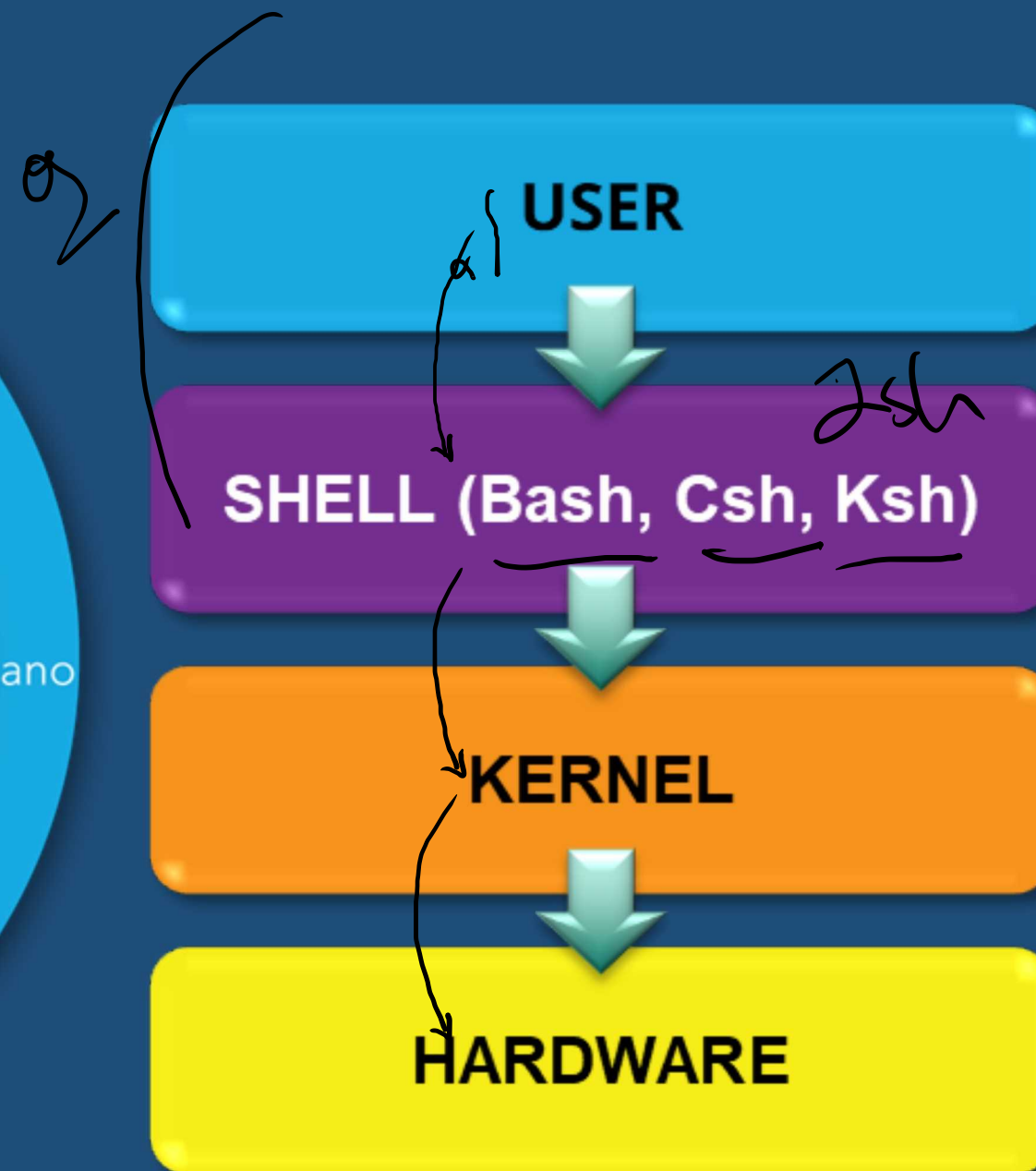
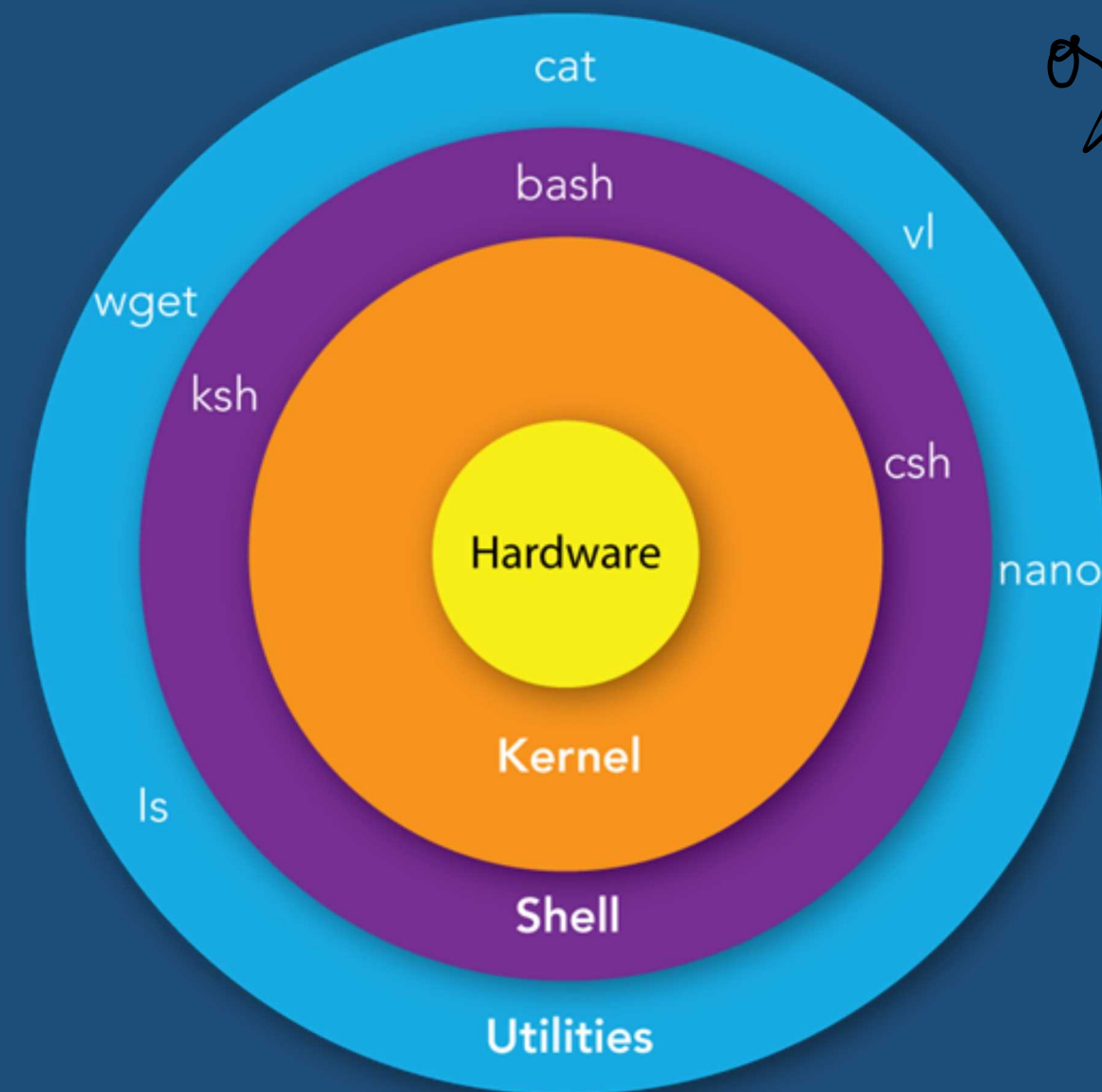
Linux Graphical Desktops

- The most popular Linux desktops: Gnome, KDE, XFCE, Cinnamon and
- Gnome has a user-friendly environment with the slogan Keep it Simple Stupid.
- KDE has a more complex, professional environment with more software.
- XFCE and Cinnamon have similar environments to the Windows Desktop.
- Ubuntu, Red Hat, SuSE and many other distributions have the default Gnome desktop.

ubuntu
└─ Gnome
└─ KDE
└─ XFCE



Shell Introduction



Ubuntu Installation Process

